ARNAV BANERJEE

BACKEND SOFTWARE ENGINEER

Bengaluru, India
github.com/xerexcoded | linkedin.com/in/arnav-banerjee

Backend software engineer with 3+ years building production financial systems, Python data workflows, and APIs. Experienced in transactional processing, SQLAlchemy, PostgreSQL, Kafka, and containerized delivery; independently build and maintain Python tools and small-business applications.

TECHNICAL SKILLS

Backend: Python, SQL, Java, FastAPI, SQLAlchemy, Spring Boot, REST APIs, OpenAPI
Data: PostgreSQL/PostGIS, Oracle, SQLite, Redis, Kafka; transactions and idempotency
Delivery: Docker, Kubernetes/OpenShift, Helm, Harness, Linux, CI/CD, Splunk observability
Cloud & AI: AWS EC2, S3, RDS (personal projects); LLM integration, RAG

PROFESSIONAL EXPERIENCE

Wells Fargo
Bengaluru, India

Software Engineer - Commercial Card Payments — Sep 2024 - Present

- Build Java/Spring Boot services and a unified API across commercial-card payments and treasury systems, including tokenization and Kafka event flows; collaborate on API contracts with product and consuming teams.
- Migrated scheduled payment jobs from Java to Python/SQLAlchemy, preserving transactions and idempotency through stored procedures and unique constraints; verified results through parallel runs and production checks.
- Modernized a reporting service producing **30,000+ reports daily**, adding shared-parameter caching and migrating template definitions from XML to YAML.
- Deliver services on OpenShift with Helm/Harness and Splunk monitoring; mentored two interns through Java 17 and Spring Boot 3 upgrades, testing, and production release.

Program Associate - Full Stack Engineer — Aug 2023 - Sep 2024

- Built Java/React features with Elasticsearch search and Redis caching; introduced chaos testing across **100+ API endpoints** to identify and resolve failure paths.
- Developed Python data-cleaning and modeling pipelines for WiDS 2024; ranked **3rd globally and 1st at Wells Fargo**.

Intern Analyst — Jan 2023 - Jul 2023

- Migrated Perl automation to Python, built QA test-data tooling, and supported an Oracle-to-CosmosDB migration with data validation.

INDEPENDENT PROJECTS

PPLX CLI | Python, Typer, SQLite, RAG — github.com/xerexcoded/pplx-cli
Build and maintain a Python research CLI on PyPI with retrieval and Markdown knowledge-graph workflows. Own tests, documentation, packaging, releases, and user issue resolution.

Small-business backends | Python, FastAPI, React
Built a clinic scheduler for 5-6 staff with availability checks and reminders, plus an Andaman Seaweed Company inventory app for two users tracking rafts, sites, and species.

Ring of Fire | PostgreSQL/PostGIS, Clojure, Next.js — kachal.ai/trivia/ringoffire
Shipped a live data application with embedded Metabase analytics; operate containerized deployments on a self-managed NixOS VPS.

EDUCATION & RECOGNITION

B.Tech, Computer Science & Engineering | NIT Durgapur | 2019 - 2023
Manager Spotlight Awards (2024, 2025) for resilience engineering, migrations, and technical teaching.